Sahaj Bhadja

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Summary

Software engineer with 2+ years building Python integration services, cloud-edge workflows, simulation pipelines, and reliable automation systems through API design, cross-platform testing, debugging, technical documentation, and AI-assisted development.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Advanced NLP, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Object-Oriented Programming, Software Engineering, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – June 2026*
- Architected multi-tenant Python services across Azure SQL, PostgreSQL, and Blob Storage for 500+ users, optimizing indexed queries and integration boundaries to reduce dashboard latency from 1.2 seconds to 350 milliseconds.
- Built fault-tolerant **FastAPI orchestration** for 1,000+ monthly jobs using explicit states, retries, validation, and failure isolation, tripling throughput while improving production troubleshooting, observability, and reliability.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Engineered Python APIs and React workflows handling 10,000+ daily requests, integrating secure document ingestion, OCR, and AWS RDS persistence while maintaining stable performance across automated production workflows.
- Designed multithreaded batch processing for 1,000+ files with automated validation, fault-tolerant execution, and failure recovery, reducing turnaround by 60% while strengthening testing, debugging, and end-user reliability.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built a Decision Transformer environment for autonomous-driving simulation, processing 100,000+ trajectory logs and improving long-horizon planning by 18% through reproducible data pipelines and systematic performance evaluation.
- Architected a spatial-temporal encoder combining HD-map geometry and historical trajectories through multi-head attention, improving closed-loop simulation stability by 15% across complex physical-world scenarios and autonomous workflows.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered **cross-platform Python microservices** and 20+ PyTest and Mockito suites through Jenkins CI/CD, automating unit and integration validation across Windows, Linux, and macOS production environments.
- Optimized API and ETL integration components using object-oriented patterns, AWS Glue, S3, and SQL, reducing response times by 30% and improving distributed data-processing efficiency by 68%.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Developed a modular platform with REST APIs, WebSockets, and eight Node.js services, supporting 2M+ synchronized records while translating complex interface requirements into maintainable real-time integration workflows.
- Automated Docker-based builds, testing, and deployments through Jenkins CI/CD, reducing release time by 50% while improving code quality, integration documentation, and defect resolution across production releases.

Skills

Languages & Integration: Python, C++, Java, JavaScript, TypeScript, SQL, Bash, Linux, object-oriented programming, design patterns, REST APIs, JSON, WebSockets, FastAPI, Flask, Node.js, microservices, interface design, cloud-edge workflows; **Automation & Data:** automation workflows, simulation, concurrent processing, PostgreSQL, SQL Server, Azure SQL, MySQL, MongoDB, AWS RDS, ETL, data pipelines, machine learning, PyTorch, Transformers, OCR, LLM orchestration; **Quality & Delivery:** AWS, Azure, GCP, Docker, Kubernetes, Jenkins, Git, CI/CD, PyTest, Mockito, unit testing, integration testing, regression testing, debugging, troubleshooting, performance optimization, monitoring, fault tolerance, API documentation, technical support, Agile/Scrum, AI-assisted development, cross-functional collaboration

Projects

CPU-Scheduling Simulator Live | *C++, React.js, JavaScript, Algorithms*
- Implemented six scheduling algorithms with dynamic metrics and Gantt visualization, validating preemptive and non-preemptive execution across 1,000+ process combinations while reducing integration and algorithm-debugging time by 20%.

HMIS Live | *TypeScript, MERN, REST APIs, Sentry, Twilio*
- Built a secure workflow platform integrating consent-based APIs, scheduling, automated alerts, and Sentry monitoring, reducing administrative workload by 30% across end-user operations and support processes.

Big-Brain Live | *Python, LLaMA-2, FAISS, Pinecone, PostgreSQL*
- Built an AI data platform for 200+ users with secure ingestion, vector indexing, semantic retrieval, and collaborative workflows, maintaining 95% query accuracy across 50,000+ experimental documents.